Skill Week :-3

Task :-1

Code:-

```c
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    char command[100];

    while (1)
    {
        // Display prompt
        printf("OSSP> ");
        fflush(stdout);

        // Read user input
        if (fgets(command, sizeof(command), stdin) == NULL)
        {
            printf("\n");
            break;
        }

        // Remove newline character
        command[strcspn(command, "\n")] = '\0';

        // Handle empty input
        if (strlen(command) == 0)
        {
            continue;
        }

        // Handle exit condition
        if (strcmp(command, "exit") == 0)
        {
```

```c
            break;
        }

        // Display entered command
        printf("You entered: %s\n", command);
    }

    return 0;
}
```

Output:-

```
anuan@Anushka:~$ nano main_loop.c
anuan@Anushka:~$ gcc main_loop.c -o main_loop
anuan@Anushka:~$ ./main_loop
OSSP> hello
You entered: hello
OSSP> ls
You entered: ls
OSSP> pwd
You entered: pwd
OSSP> test
You entered: test
OSSP> exit
Exiting shell...
```

Task:-2

Code:-

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int fd1, fd2;
    char buffer[100];
    ssize_t n;

    // Open File1.txt for reading
    fd1 = open("File1.txt", O_RDONLY);

    if (fd1 < 0)
    {
        perror("Error opening File1.txt");
        return 1;
    }

    // Open File2.txt for writing
    // Create it if it does not exist
    // Truncate it if it already exists
    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (fd2 < 0)
    {
        if (write(fd2, buffer, n) != n)
        {
            perror("Error writing to File2.txt");
            close(fd1);
            close(fd2);
            return 1;
        }
    }

    if (n < 0)
    {
        perror("Error reading File1.txt");
    }
    else
    {
        printf("File copied successfully!\n");
    }

    // Close both files
    close(fd1);
    close(fd2);

    return 0;
```

Output:-

```
anuan@Anushka:~$ nano File1.txt
anuan@Anushka:~$ nano file_copy.c
anuan@Anushka:~$ gcc file_copy.c -o file_copy
anuan@Anushka:~$ ./file_copy
File copied successfully!
anuan@Anushka:~$ ls
File1.txt   file_copy.c        hello        process.c
File2.txt   file_copy.c.save   hello.c      process.c.save
OSSP        fork_exec          main_loop    program.c
Practical   fork_exec.c        main_loop.c  pwd
Skill       fork_exec.c.save   process
file_copy   fork_exec.r        process.
anuan@Anushka:~$ cat Filed.txt
cat: Filed.txt: No such file or directory
anuan@Anushka:~$ cat File1.txt
Hello, this is File1.
This content is copied using Linux system calls.
Open, Read, Write and Close are used.
anuan@Anushka:~$ cat File2.txt
Hello, this is File1.
This content is copied using Linux system calls.
Open, Read, Write and Close are used.
anuan@Anushka:~$ |
```

Task :-3

Code:-

```c
#include <stdio.h>
#include <string.h>

#define MAX 100

int main()
{
    char buffer[MAX];
    int index = 0;
    char ch;

    printf("Simple Keyboard Input Program\n");
    printf("Type a command (type 'exit' to quit)\n");

    while (1)
    {
        printf("OSSP> ");
        index = 0;

        while (1)
        {
            ch = getchar();

            // Enter key
            if (ch == '\n')
            {
```

```c
                    printf("\b \b");
                }
            }

            // Store normal characters
            else
            {
                if (index < MAX - 1)
                {
                    buffer[index++] = ch;
                }
            }
        }

        if (strcmp(buffer, "exit") == 0)
        {
            printf("Exiting Program...\n");
            break;
        }

        printf("Command Entered: %s\n", buffer);
    }

    return 0;
}
```

Output:-

```
anuan@Anushka:~$ ./keyboard_a
Simple Keyboard Input Program
Type a command (type 'exit' to quit)
OSSP> hello
Command Entered: hello
OSSP> operating system
Command Entered: operating system
OSSP> linux
Command Entered: linux
OSSP> exit
Exiting Program...
anuan@Anushka:~$
```